THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Planaria	Cut pieces regrow
2	**(a)** ovary	Fruit
3	**(d)** anther	Stamen
4	**(a)** Correct	Both
5	**(b)** stem cutting	Nodes
6	**(c)** phenotype	Observable
7	**(d)** 3 : 1	Phenotypic
8	**(d)** alleles	T and t
9	**(a)** Correct	Haploid
10	**(a)** 1/2	50%

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gamete: reproductive cell with half chromosomes; zygote: formed by fusion of two gametes, full chromosome number.	2
12	Rose, jasmine, mango, orange (any two)	1+1
13	A checkerboard used to show the possible combinations of gametes and the offspring in a cross.	2
14	All tall (Tt).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Creates variation; helps survival in changing environments; combination of DNA from two parents leads to evolution	3
16	Genes code for proteins/enzymes; e.g. the gene for an enzyme that makes a plant growth hormone — efficient enzyme gives tall plant, inefficient gives short	3
17	(i) carries egg; site of fertilisation (ii) implantation and development of embryo (iii) nutrition, gas exchange and waste removal for embryo	1+1+1
	OR: An outgrowth (bud) develops due to repeated cell division, grows into a tiny individual and detaches to live independently	3
18	TT : Tt : tt = 1 : 2 : 1; tall : short = 3 : 1	1½+1½
	OR: Inherited: passed through genes, present from birth (eye colour); acquired: developed during life, not passed (learning to swim)	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) sepals, petals, stamens, pistil (ii) self: same flower; cross: different flower of the same species (iii) flower with either stamens or pistil (papaya)	2+2+1
	OR: (i) vegetative propagation (ii) faster, identical to parent (iii) bending a branch to soil to root before separation (iv) they are seedless (v) jasmine/sugarcane	1+1+1+1+1
20	(i) crossed pure tall with pure short and self-pollinated F1 (ii) F1 all Tt (tall); F2: 1 TT : 2 Tt : 1 tt; 3 tall : 1 short (iii) traits are controlled by factors; one is dominant; factors segregate	1+2+2
	OR: (i) RY, Ry, rY, ry (ii) 9 : 3 : 3 : 1 (iii) 1 (iv) independent assortment (v) 12/16 = 3/4	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both are correct	Both functions
2	**(a)** hibiscus	Both stamens and pistil
3	**(c)** kala-azar	Protozoan
4	**(a)** vasectomy	Surgical method
5	**(b)** thick wall	Resistant
6	**(a)** Student A is correct	Mendel
7	**(b)** tt	Recessive needs two
8	**(d)** 4	9 : 3 : 3 : 1
9	**(c)** 44	22 pairs
10	**(b)** shorter plants	Less hormone

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Seedless plants can be grown; plants mature faster; identical to parent.	2
12	False — Amoeba reproduces by binary fission; Plasmodium by multiple fission.	2
13	Gregor Johann Mendel; garden pea	1+1
14	False — they do not change the DNA of germ cells, so they are not inherited.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zygote divides to form embryo; ovule → seed; ovary → fruit; petals, sepals, stamens wither	3
16	Mother gives only X; father gives X or Y; the type of sperm fusing with the egg decides the sex	3
17	Sexually transmitted diseases: gonorrhoea, HIV-AIDS; use condoms, avoid multiple partners	1+1+1
	OR: Body's energy goes to general growth; at puberty: hair growth, oily skin, pimples, genital changes	3
18	Both Tt: Tt × Tt → TT, Tt, Tt, tt (1/4 short)	3
	OR: Used pure lines; studied one trait at a time; used large numbers and statistical analysis	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) testes (sperm and testosterone), scrotum (lower temperature), vas deferens, seminal vesicles, prostate, urethra, penis (ii) testosterone controls development of secondary sexual characters	4+1
	OR: (i) fission: Amoeba; fragmentation: Spirogyra; regeneration: Planaria (ii) they have specialised cells organised into tissues and organs	3+2
20	(i) XX mother gives X eggs; XY father gives X or Y sperm; X + X → girl; X + Y → boy (ii) the father's sperm determines the sex	3+2
	OR: (i) inheritance of traits (ii) genes: DNA segments; chromosomes: thread-like structures carrying genes (iii) gene → protein → trait (iv) free earlobe/dark eyes	1+2+1+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** zygote	Diploid cell
2	**(a)** Correct	Hormonal method
3	**(b)** Spirogyra	Filamentous alga
4	**(b)** seminal vesicles and prostate	Semen
5	**(a)** Student B is correct	Not 100% accurate
6	**(a)** males only	XY
7	**(b)** Incorrect — all are tall	TT, Tt only
8	**(c)** seven	Height, colour, shape…
9	**(c)** fertilisation	Zygote
10	**(a)** Student A is correct	Test cross

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Produces eggs (ova) and secretes hormones (oestrogen).	2
12	If the egg is not fertilised, the thick uterine lining is not needed and breaks down.	2
13	They inherit genes from both parents in new combinations.	2
14	A plant that on self-pollination always produces offspring with the same trait (homozygous).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nucleus divides repeatedly within a cyst; each nucleus surrounded by cytoplasm forms many daughter cells released when the cyst bursts	3
16	Four: TR, Tr, tR, tr	1+2
17	Wind (maize), insects (sunflower), water (vallisneria), birds	3
	OR: Gametes have half the chromosomes (formed by special division); their fusion restores the full number in the zygote	3
18	1/2 — half the sperms carry X and half carry Y	3
	OR: Deoxyribonucleic acid; in chromosomes in nucleus; carries hereditary information for making proteins	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) placenta (ii) finger-like projections on the embryo side of placenta (iii) glucose, oxygen (iv) wastes (v) about nine months	1+1+1+1+1
	OR: (i) barrier, hormonal, IUCD, surgical (ii) barrier prevents sperms meeting egg; pills prevent egg release; copper-T prevents implantation; surgery blocks tubes (iii) to prevent unwanted pregnancy, STDs and control population	2+2+1
20	(i) violet (ii) 3 violet : 1 white (iii) VV and Vv (iv) cross it with white (vv): pure gives all violet (v) vv	1+1+1+1+1
	OR: (i) RRYY × rryy → RrYy; F2 16 boxes: 9 : 3 : 3 : 1 (ii) parental: round yellow, wrinkled green; new: round green, wrinkled yellow	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** style	Pistil part
2	**(d)** stem tuber (eyes)	Buds on tuber
3	**(a)** 2 to 8 days	Monthly cycle
4	**(a)** 46	Diploid
5	**(d)** cross-pollination	Agents: wind, water, insects
6	**(d)** provides information for one protein	Codes protein
7	**(b)** 4	RY, Ry, rY, ry
8	**(b)** XX	Female
9	**(a)** inherited traits	Heredity
10	**(c)** 3	Rryy/RRyy

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Stigma, style, ovary	2
12	They store food for the developing embryo.	2
13	Sperms carrying X chromosome and sperms carrying Y chromosome.	2
14	The two alleles of a gene separate during gamete formation, so each gamete gets one.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Can cause infection if not done properly; are difficult to reverse; misused for sex-selective abortion	3
16	In a dihybrid cross, F2 showed new combinations (round green, wrinkled yellow), so the inheritance of seed shape and colour was independent	3
17	Many disease-free plants from one plant quickly; used for orchids, ornamentals; no seeds needed	3
	OR: Zygote would have double the chromosomes; chromosome number would double every generation, disturbing DNA balance	3
18	F1 is heterozygous (Tt); in F2 two recessive alleles can combine (tt), so the recessive trait is expressed	3
	OR: One copy is inherited from each parent; allows dominance/recessive expression and variation	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) helps survival under changing conditions (ii) errors in DNA copying; combination of DNA in sexual reproduction (iii) different heights/eye colour	2+2+1
	OR: (i) ovaries, oviducts, uterus, cervix, vagina (ii) one egg released monthly; uterine wall thickens; without fertilisation lining breaks down — menstruation	3+2
20	(i) dominant expressed even in one copy; recessive only in two copies (ii) TT × tt → Tt all tall; t is masked by T	2+3
	OR: (i) combination of genes from two parents and DNA copying errors (ii) survival in changing environment (iii) heat-resistant bacteria survive a heat wave	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.